ASPEED Technology Company Profile

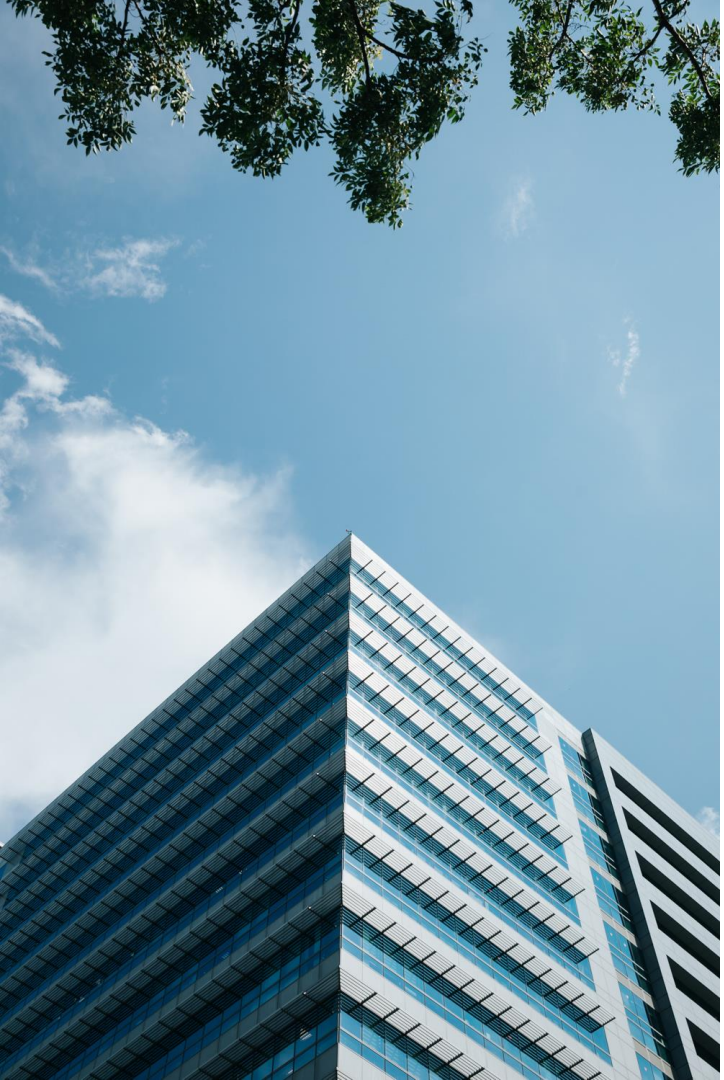
Q4, 2023

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Safer Harbor Notice

- ASPEED's statements of its current expectations are forward-looking statements subject to significant risks and uncertainties and actual results may differ materially from those contained in the forward-looking statements. Except as required by law, we undertake no obligation to update any forward-looking statement, whether as a result of new information, future events, or otherwise.

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About ASPEED

- Founded in 2004
- Fabless-lite IC design company
- Number of employees: 125 (as of Nov. 2023)
- Locations:
  - Headquarters: Hsinchu City, Taiwan
  - US office: San Jose, CA
- Product lineup:
  - Cloud & Enterprise (BMC/BIC/PFR)
  - Smart AV (Cupola360 & AVoIP SoC)

Vision & Value



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Corporate Management Team

Chairman & President
Chris Lin



VP of Sales
Luke Chen



COO
CJ Hsieh



VP of R&D
Hungju Huang



US Country Head
Dwarka Partani



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ASPEED

Milestones

2004.11

- ASPEED founded

2005-2006

- Introduced 1st gen BMC SoC AST2000/1000
- Delivered 1st mass production of SoC shipment

2007-2008

- ISO 9001:2000 certified
- Introduced 2nd gen BMC SoC AST2100/2050/1100
- Introduced 3rd gen BMC SoC AST2200/2150

2009-2011

- Launched new product line: PC/AV Extension SoC
- Received funding from Intel Capital

2012-2014

- Listed on the Taiwan OTC stock market
- Listed on Taipei Exchange
- Introduced 6th gen BMC SoC AST2500

2015-2016

- BMC SoC market share ranked No.1
- Acquired Broadcom Emulex Pilot™ BMC SoC Business
- ISO9001:2015 certified

2017-2018

- Joined Open Compute Project (OPC)
- Introduced new product line: Cupola360 Spherical Image Processor
- ISO 14001:2015 certified

2019-2023

- Introduced 7th gen 28nm BMC SoC AST2600
- ISO 27001:2015 certified
- ISO 14064:2018 certified

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Awards & Recognition

2014 - 2023

- Forbes Asia's Best Under A Billion for **10** consecutive years

2009

- COMPUTEX Best Choice Award

2008

- Ranked 37th in Deloitte Asia Pacific Technology Fast 500

2020

- Lenovo Technology Innovation Award
- Chairman & President Mr. Chris Lin received NYCU Distinguished Alumni Award



2021

- Chairman & President Mr. Chris Lin received Outstanding ICT Elite Award and NTHU Distinguished Alumni Award
- High-Growth Companies Asia Pacific 2021 from the Financial Times



2022

- COO Mr. CJ Hsieh received SEMI Industry Contribution Award and ISES Advisory Board Member Award
- Taiwan Best-in-Class 100 from Taiwan Institute of Directors and CDRC Consulting Group



2023

- Chairman & President Mr. Chris Lin received the ERSO Award
- InfoComm Best of Show
- Ranked 1st in Most Honored Company (All Asia Executive Team – Semiconductor) from Institutional Investor
- Ranked 5th in Business Performance of Taiwan's Top 5000 Large Enterprises from CRIF

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Remarkable Highlights

2022 Market Cap
(NTD)

63.7 Billion

10Y CAGR
(2013-2022)



27.4%

BMC SoC Market Share

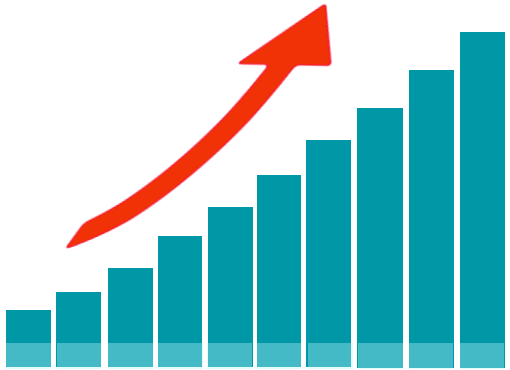


World's No.1

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Corporate Resilience

Achieved record revenue and profit since founding for **18** consecutive years



"Forbes Asia's Best Under A Billion" for **10** consecutive years



Forbes Asia

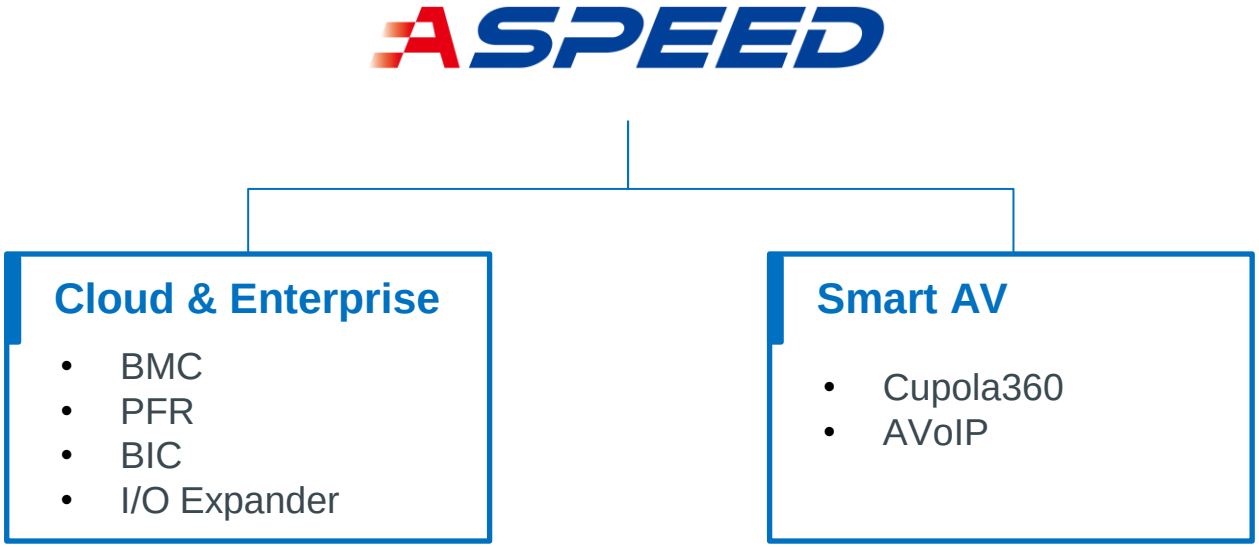
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Product & Technology

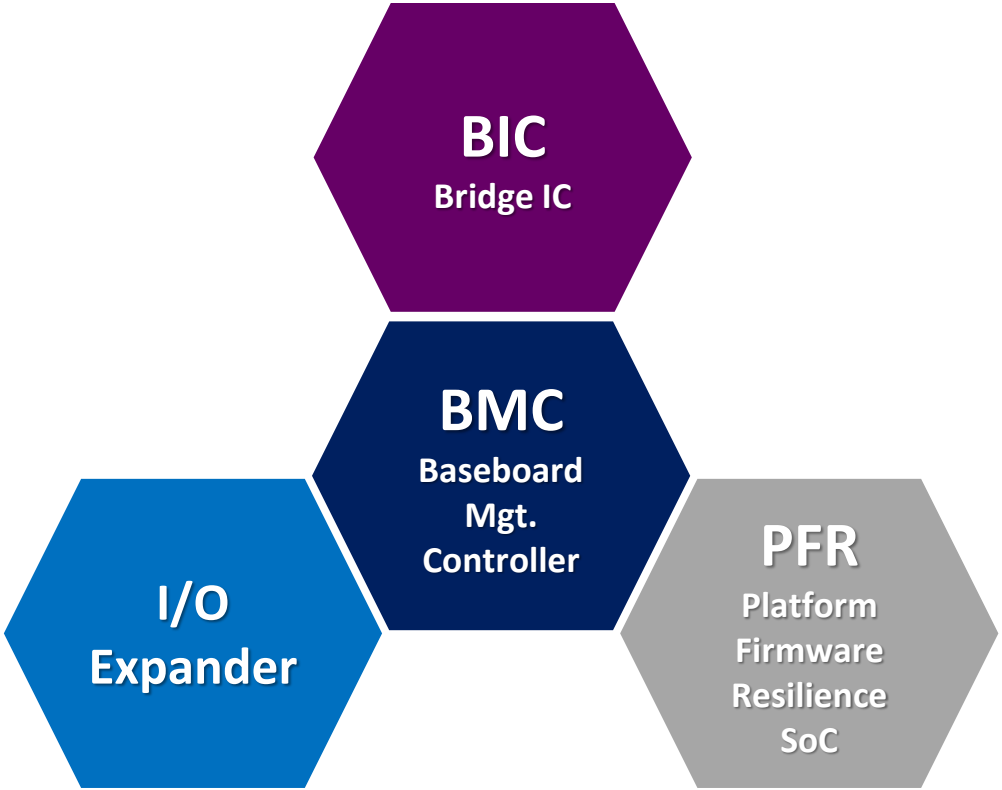
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Product Portfolio



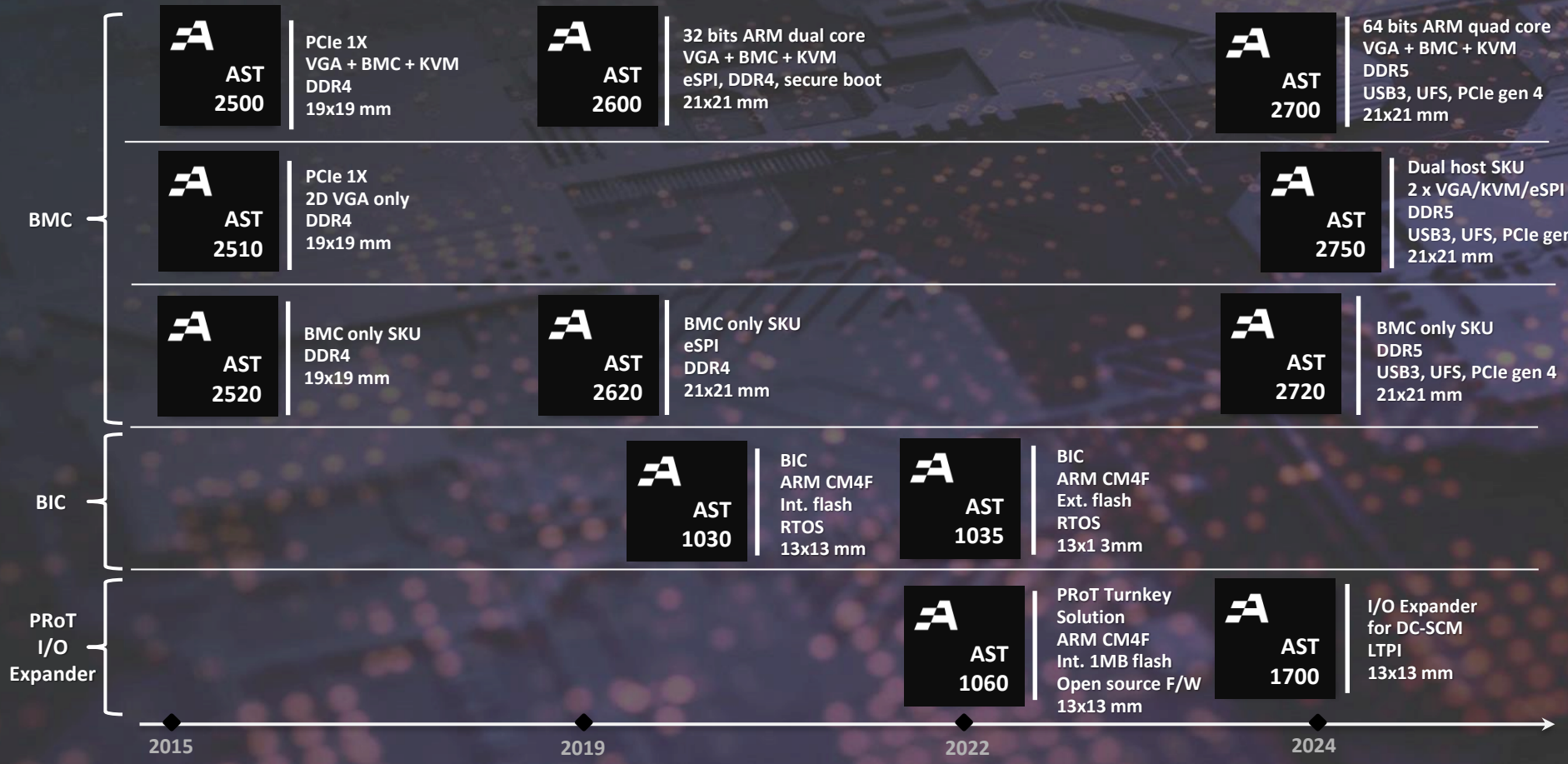
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Cloud & Enterprise Solutions



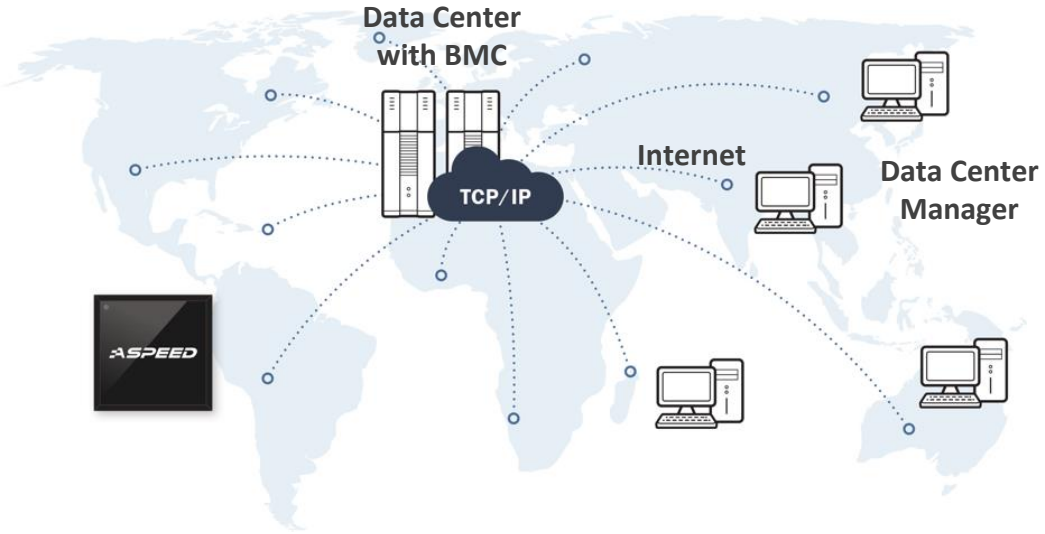
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Cloud & Enterprise Roadmap



Baseboard Management Controller (BMC)

- Specialized processor includes 2D graphics and logic control features
- **Used for remote monitoring and management of host systems**
- Located on the motherboard of **computers, servers, networks, and storage devices**
- Can be **accessed remotely via dedicated or shared networks**
- Monitor hardware via sensors, flash BIOS/UEFI
- Provide host access via serial or physical/virtual KVM Console, and power cycle to record host and log events
- **Streamlines operations, saving time and manpower** by eliminating the need for administrators to physically inspect each server in the rack while still enabling them to perform maintenance.

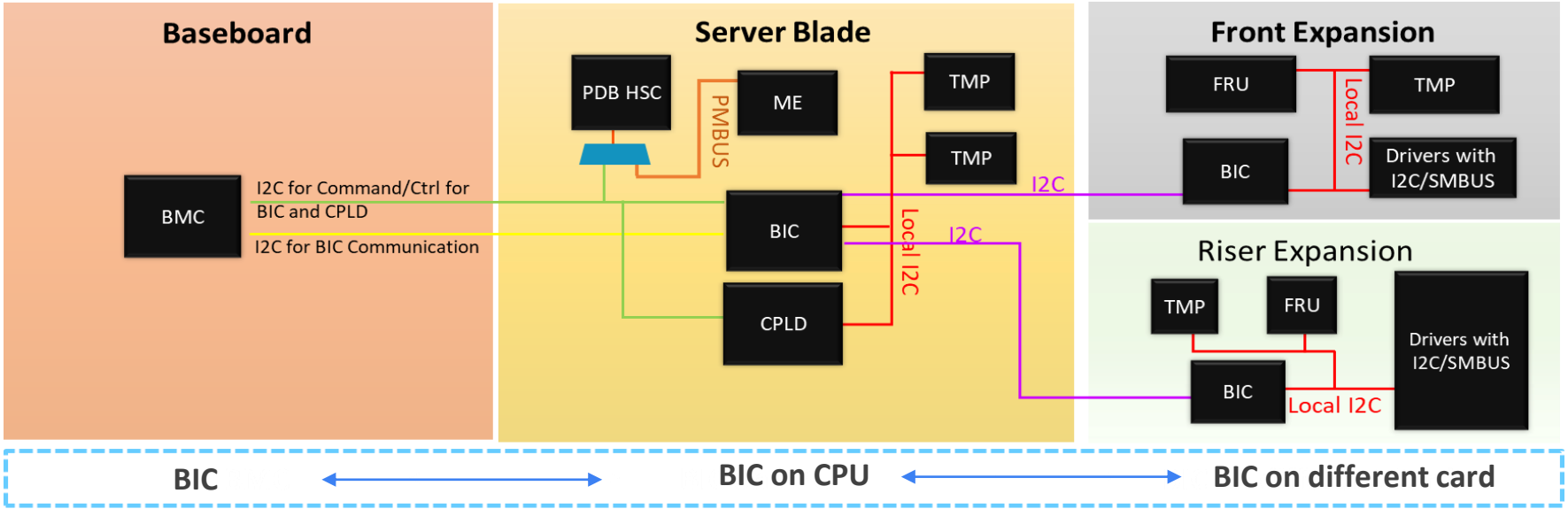


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Bridge IC (BIC)

- **Compact processor equipped with integrated memory and storage**
- **In tandem with BMC** within OCP framework, **enhancing its management capabilities**

- **Improve operational efficiency** by acting as a localized BMC agent
- **Features built-in logic control** and **supports multiple connections to the host system and BMC**, enabling event logging from the system where it is deployed



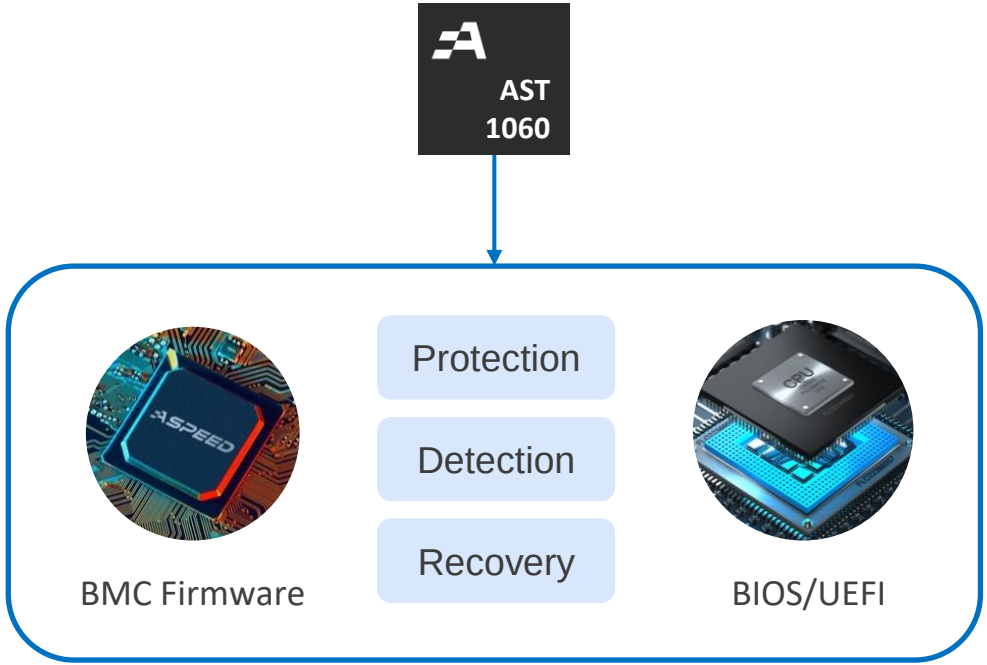
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Platform Firmware Resilience (PFR) SoC

ASPEED utilizes the PFR SoC to ensure **firmware security across the enterprise platform**, effectively safeguarding the platform firmware against potential malicious attacks.

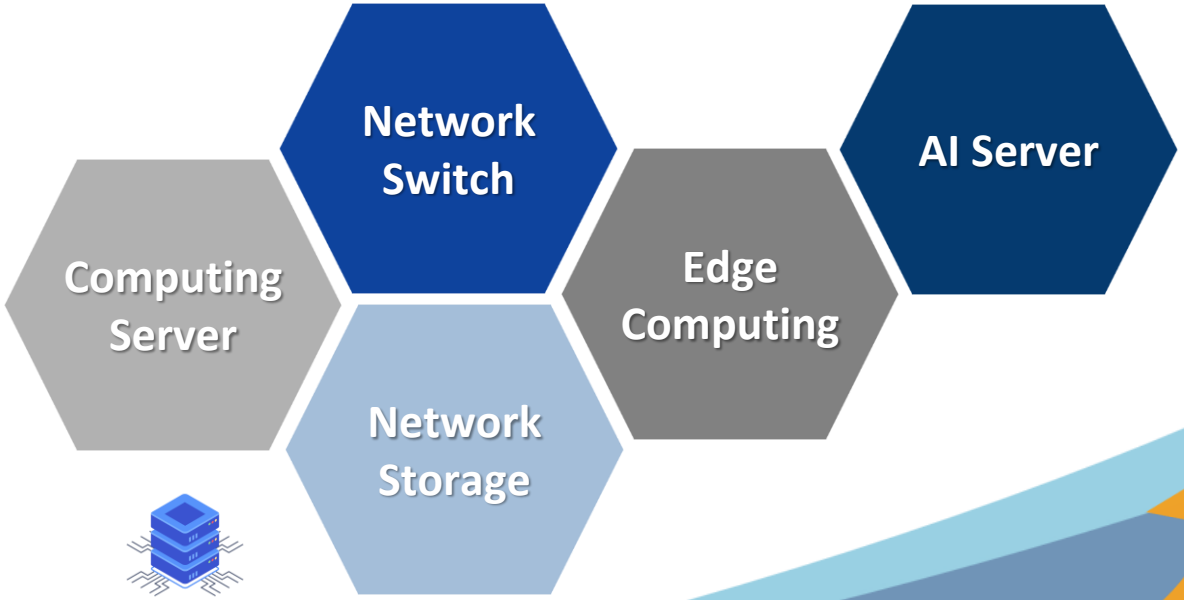
PFR Guideline

- Published by National Institute of Standards and Technology (NIST) in 2018
- A security specification intended to **protect platform firmware, system BIOS/UEFI, and BMC firmware from malicious attacks**
- Firmware must meet three criteria: **protection, detection and recovery**



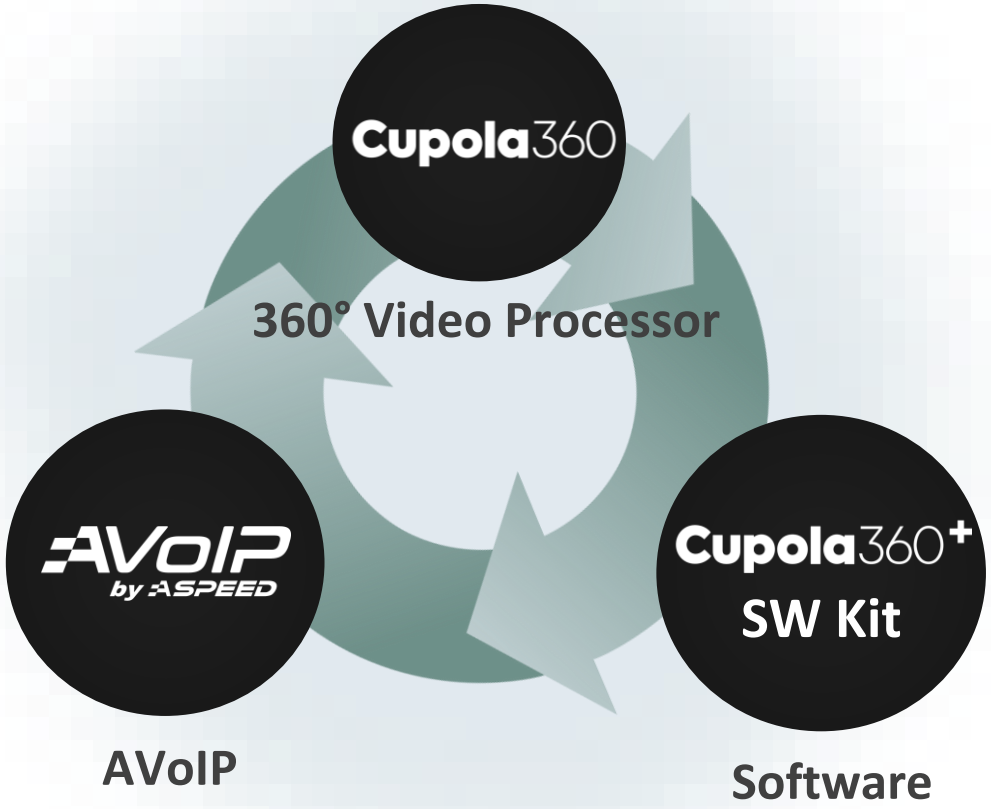
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Cloud & Enterprise Applications



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Smart AV Solutions



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Smart AV Solutions

AVoIP by ASPEED

AST 1520
4K2K@30
Matrix AV over IP Solutions
456-pin 19x19 mm, TFBGA

AST 1220
4K2K@30
Image Stitching Processor
USB Webcam
500-pin 12x12 mm, TFBGA

Cupola360

- Video Conference

AST 1620
4K2K@30
1 to 1 AV over IP
USB Webcam over IP
128-pin 14x14 mm
LQFP



AST 1230
8K2K@30
Image Stitching Processor
Hybrid Video Conferencing
426-pin 13x13 mm, CSP

- Video Wall
- Digital Signage
- Command Center

AST 1630
4K2K@60
1 to Many AV over IP
400-pin 17x17 mm
TFBGA

- ImmersiveX
- LIVE 3D Experience



AST 1530 ⟷ **AST 1535**
Two-chip solutions for
4K2K@60
Matrix AV over IP

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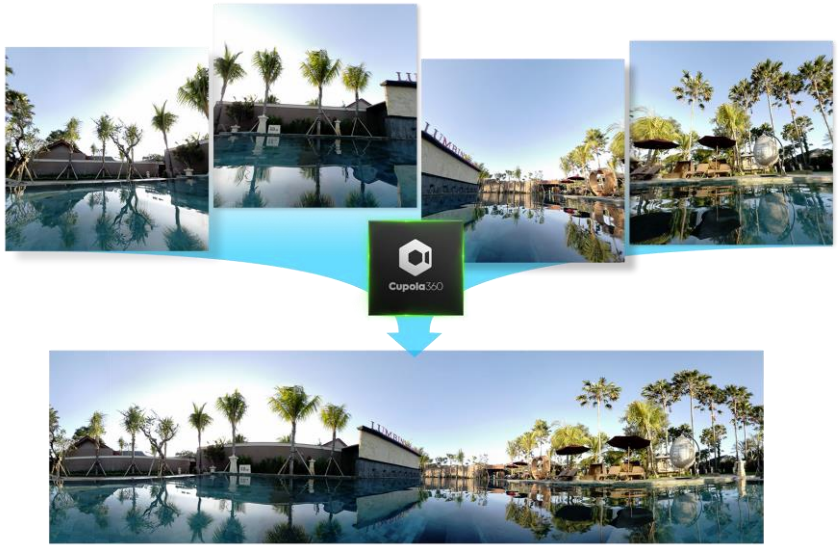
ASPEED

Cupola360 Spherical Image Stitching Solution

- Designed with proprietary **in-camera image stitching technology tailored for 360-degree images**
- **Real-time and optically perfect panoramic views with ultra-high resolution**

- Cupola360 offers an **efficient and optimized 360-degree camera total solution for real-time immersive scenarios with its self-developed Cupola360⁺ Software Kit.**

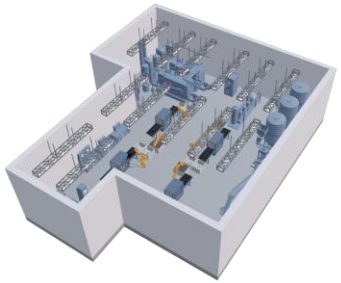
- **Application scenarios:**
  - Video conferencing
  - Remote inspection and audit
  - Virtual factory tours
  - Education
  - … and many more



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Real-time Immersive Experience Powered by Cupola360

3D Model + 360 Camera + AI =



3D & AI Modelling



Cupola360
360-degree Camera



Deep Learning



Live 3D
Immersive Experience

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ASPEED

AV over IP Extension (AVoIP)

ASPEED's AVoIP SoC employs standard IPs featuring a dual-codec engine and proprietary compression technology, enabling support for **long-distance audio/video extension** at up to **4K60 visual lossless quality** using standard **ethernet** cables.



Application Scenarios

- Command centers
- Control rooms
- Conference rooms
- Lecture halls
- Court rooms
- Sports bars

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Financial Highlights

Q3, 2023

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Q3’23 Income Statement

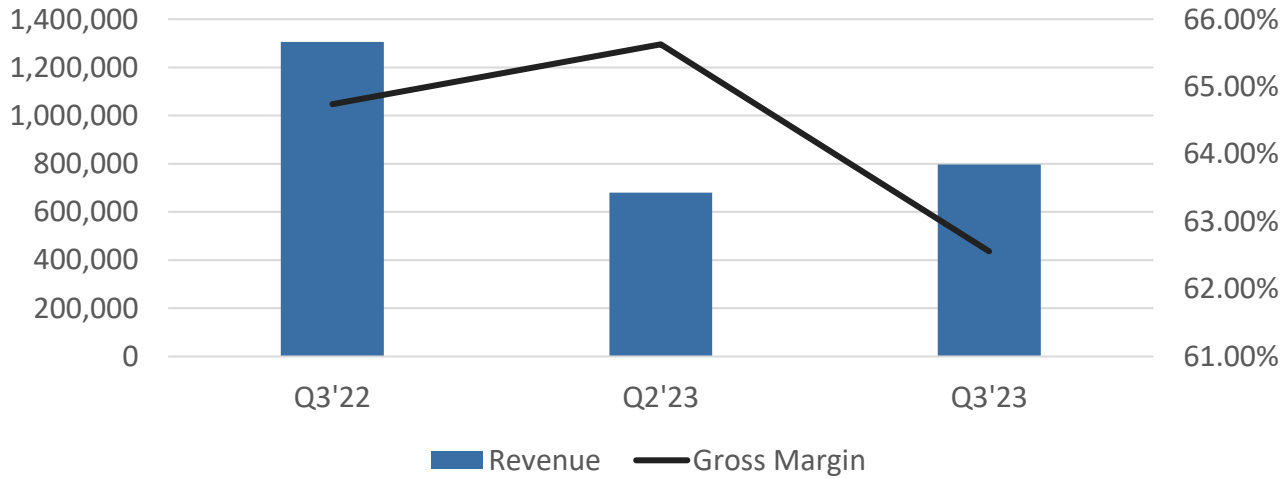
Unit : NTD 1,000

	Q1’23		Q2’23		Q3’23		Q3’23 over Q2’23	Q3’23 over Q3’22
	NTD	%	NTD	%	NTD	%		
Operating Revenue	669,396	100.00%	680,039	100.00%	796,713	100.00%	17.16%	-39.02%
Gross Profit	429,994	64.24%	446,316	65.63%	498,395	62.56%	11.67%	-41.08%
Operating Expenses	220,549	32.95%	231,377	34.02%	235,186	29.52%	1.65%	-7.14%
Operating profit	209,445	31.29%	214,939	31.61%	263,209	33.04%	22.46%	-55.58%
Non-operating income and expenses	20,004	2.99%	68,374	10.05%	76,338	9.58%	11.65%	-50.28%
Pre-Tax Profit	229,449	34.28%	283,313	41.66%	339,547	42.62%	19.85%	-54.49%
Net Income	186,309	27.83%	240,352	35.34%	270,455	33.95%	12.52%	-54.80%
EPS (Basic)	4.93	-	6.37	-	7.16	-	12.40%	-54.77%

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Consolidated Revenue

Unit : NTD 1,000

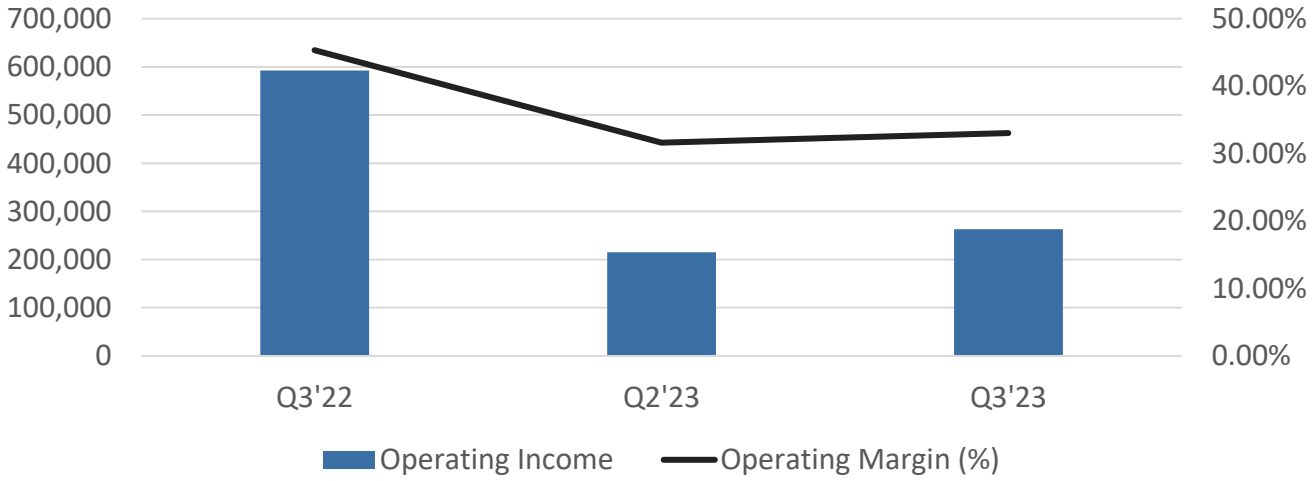


Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
Revenue	1,306,580	680,039	796,713	17.16%	-39.02%
Gross Margin	64.74%	65.63%	62.56%	-4.68%	-3.37%

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Consolidated Operating Income

Unit : NTD 1,000

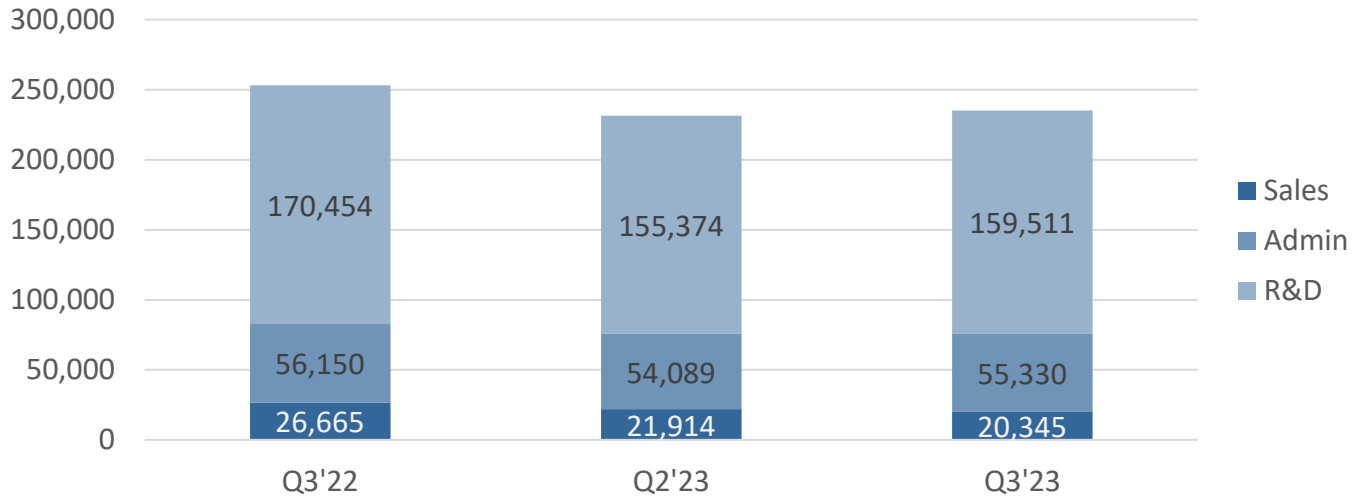


Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
Operating Income	592,568	214,939	263,209	22.46%	-55.58%
Operating Margin	45.35%	31.61%	33.04%	4.52%	-27.16%

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Consolidated Operating Expenses

Unit : NTD 1,000

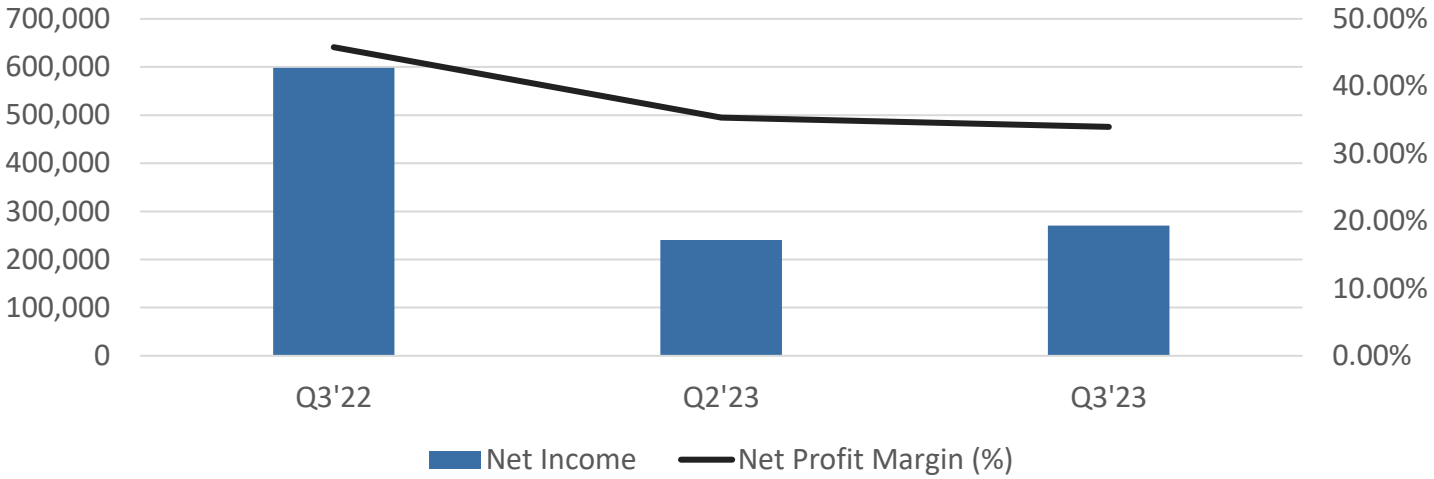


Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
Operating Expenses	253,269	231,377	235,186	1.65%	-7.14%

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Consolidated Net Income

Unit : NTD 1,000

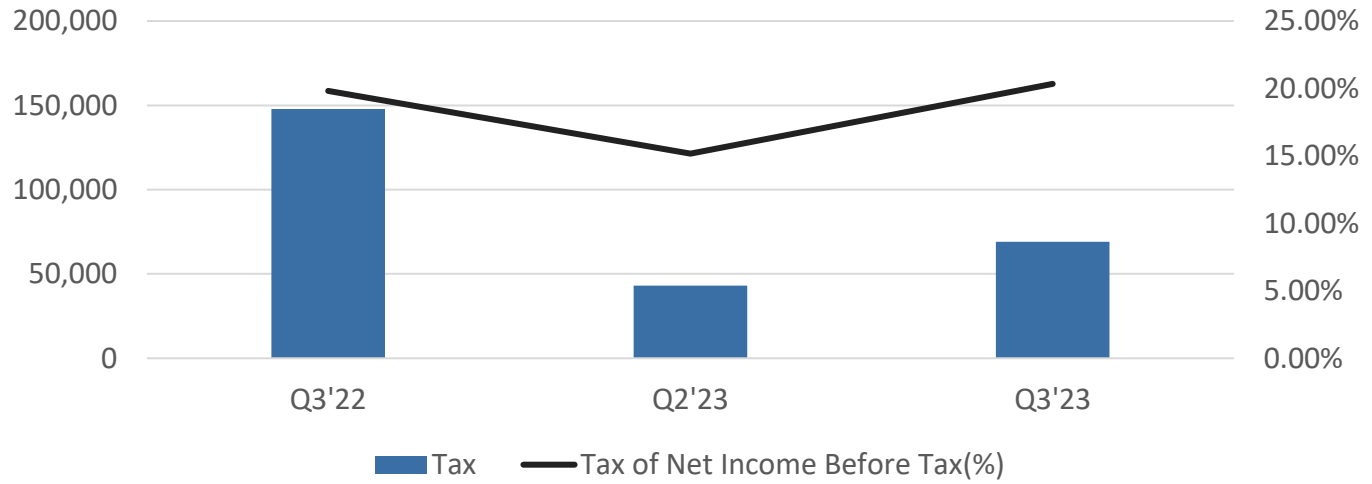


Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
Net Income	598,343	240,352	270,455	12.52%	-54.80%
Net Profit Margin	45.79%	35.34%	33.95%	-3.95%	-25.87%

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Consolidated Tax

Unit : NTD 1,000

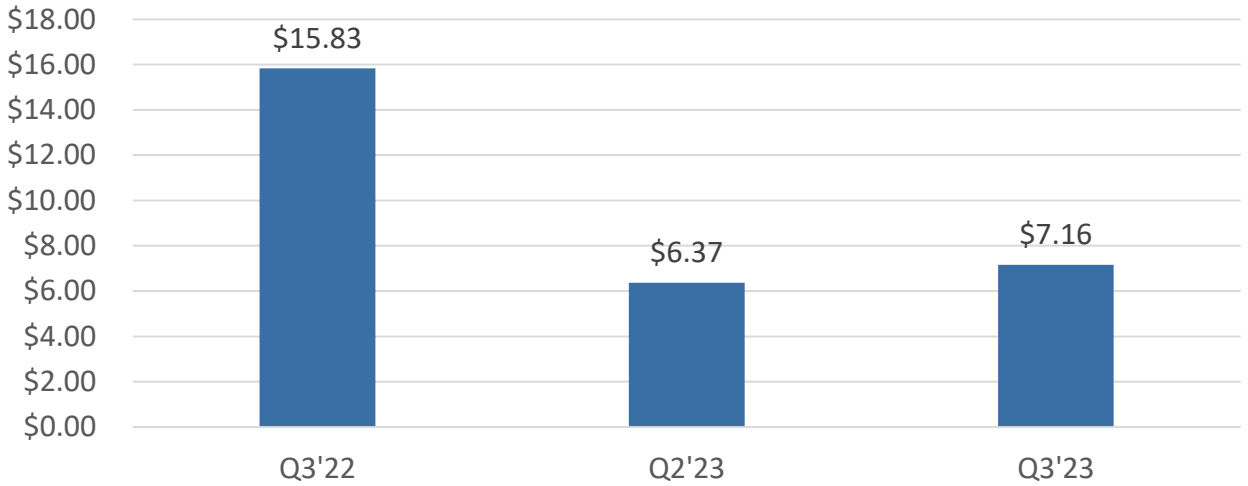


Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
Tax	147,770	42,961	69,092	60.82%	-53.24%
Tax of Net Income Before Tax	19.81%	15.16%	20.35%	34.19%	2.74%

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Consolidated Earning Per Share

Unit : NTD



Quarterly	Q3’22	Q2’23	Q3’23	QoQ	YoY
EPS	15.83	6.37	7.16	12.40%	-54.77%

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ESG Highlights

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Sustainability Strategy



Strengthen corporate governance and resilience

- Strengthening risk management strategies
- Building a complete corporate sustainable governance framework and system



Focus on green energy-saving core technologies

- Regularly performing GHG and passing third-party verification
- Power saving and carbon reduction, formulating carbon reduction targets and actions
- Strengthening core R&D and improving power consumption by chips



Build a talented sustainable technology workforce in Taiwan

- Developing core values of sustainability and promoting ESG for all employees
- Strengthening the cultivation of talents and seeking to advance women in technology industry

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ESG Highlights in 2022

Governance

- Revenue and profit grew by **43.23%** and **60.38%** respectively, reaching a record high for **18** consecutive years
- Obtained **58** global patents
- Customer satisfaction rate reached **94.74%**

Environment

- Officially signed as a **TCFD supporter** in response to the TCFD initiative
- Received **ISO14064-1: 2018 GHG inventory certification**
- In response to the SBTi SME Route, set net-zero targets and paths
- BMC AST2600 saved over **61%** of energy

Social

- Ranked **1st** in median remuneration of non-executive directors among the TWSE/TPEx listed companies
- Employee benefits welfare of NTD **522 million**, annual increase of 51.74%
- Social donation reached NTD **13.8 million**



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2022 Sustainability Report & ESG Summary

Sustainability Report



中文 English

ESG Summary



English

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